LeoChiuu/sbert-base-ja

like 0

Sentence Similarity | sentence-transformers | Safetensors | bert | feature-extraction | Generated from Trainer | dataset_size:124 | loss:MultipleNegativesRankingLoss | Eval Results | text-embeddings-inference | arxiv:1908.10084 | arxiv:1705.00652

Train | Deploy | Use this model

Model card | Files xet | Community

Downloads last month

-

Safetensors

Model size 111M params | Tensor type F32 | Files info

Inference Providers NEW

Sentence Similarity

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for LeoChiuu/sbert-base-ja

Base model	colorfulscoop/sbert-base-ja
Finetuned (5)	this model

Evaluation results

Cosine Accuracy on custom arc semantics data self-reported	0.968
Cosine Accuracy Threshold on custom arc semantics data self-reported	0.295
Cosine F1 on custom arc semantics data self-reported	0.984
Cosine F1 Threshold on custom arc semantics data self-reported	0.295
Cosine Precision on custom arc semantics data self-reported	1.000
Cosine Recall on custom arc semantics data self-reported	0.968
Cosine Ap on custom arc semantics data self-reported	1.000

Dot Accuracy on custom arc semantics data self-reported 0.968

Dot Accuracy Threshold on custom arc semantics data self-reported 144.980

Dot F1 on custom arc semantics data self-reported 0.984

Expand 35 evaluations

View on Papers With Code

SentenceTransformer based on colorfulscoop/sbert-base-ja

This is a sentence-transformers model finetuned from colorfulscoop/sbert-base-ja. It maps sentences & paragraphs to a 768-dimensional dense vector space and can be used for semantic textual similarity, semantic search, paraphrase mining, text classification, clustering, and more.

Model Details

Model Description

- **Model Type:** Sentence Transformer
- **Base model:** colorfulscoop/sbert-base-ja
- **Maximum Sequence Length:** 512 tokens
- **Output Dimensionality:** 768 tokens
- **Similarity Function:** Cosine Similarity

Model Sources

- **Documentation:** Sentence Transformers Documentation
- **Repository:** Sentence Transformers on GitHub
- **Hugging Face:** Sentence Transformers on Hugging Face

Full Model Architecture

```
SentenceTransformer(
  (0): Transformer({'max_seq_length': 512, 'do_lower_case': False}) with Transform
  (1): Pooling({'word_embedding_dimension': 768, 'pooling_mode_cls_token': False,
```

```
)
```

Usage

Direct Usage (Sentence Transformers)

First install the Sentence Transformers library:

```
pip install -U sentence-transformers
```

Then you can load this model and run inference.

```
from sentence_transformers import SentenceTransformer

# Download from the 🤗 Hub
model = SentenceTransformer("LeoChiuu/sbert-base-ja")
# Run inference
sentences = [
    'どっちも欲しくない',
    'タイマツ要らない',
    '花壇を調べよう',
]
embeddings = model.encode(sentences)
print(embeddings.shape)
# [3, 768]

# Get the similarity scores for the embeddings
similarities = model.similarity(embeddings, embeddings)
print(similarities.shape)
# [3, 3]
```

Evaluation

Metrics

Binary Classification

- Dataset: `custom-arc-semantics-data`
- Evaluated with [`BinaryClassificationEvaluator`]

Metric	Value
cosine_accuracy	0.9677
cosine_accuracy_threshold	0.2948
cosine_f1	0.9836
cosine_f1_threshold	0.2948
cosine_precision	1.0
cosine_recall	0.9677
cosine_ap	1.0
dot_accuracy	0.9677
dot_accuracy_threshold	144.9802
dot_f1	0.9836
dot_f1_threshold	144.9802
dot_precision	1.0
dot_recall	0.9677
dot_ap	1.0
manhattan_accuracy	0.9677
manhattan_accuracy_threshold	585.5504
manhattan_f1	0.9836
manhattan_f1_threshold	585.5504
manhattan_precision	1.0

Metric	Value
manhattan_recall	0.9677
manhattan_ap	1.0
euclidean_accuracy	0.9677
euclidean_accuracy_threshold	26.3433
euclidean_f1	0.9836
euclidean_f1_threshold	26.3433
euclidean_precision	1.0
euclidean_recall	0.9677
euclidean_ap	1.0
max_accuracy	0.9677
max_accuracy_threshold	585.5504
max_f1	0.9836
max_f1_threshold	585.5504
max_precision	1.0
max_recall	0.9677
max_ap	**1.0**

Training Details

Training Dataset

Unnamed Dataset

- Size: 124 training samples
- Columns: `text1`, `text2`, and `label`

- Approximate statistics based on the first 1000 samples:

	text1	text2	label
type	string	string	int
details	• min: 4 tokens • mean: 8.59 tokens • max: 14 tokens	• min: 5 tokens • mean: 8.58 tokens • max: 14 tokens	• 1: 100.00%

- Samples:

text1	text2	label
`昨晩何を食べたの？`	`昨夜何を食べたの？`	`1`
`スリッパをはいたの？`	`スリッパはいてた？`	`1`
`家の中`	`家の中へ行こう`	`1`

- Loss: [`MultipleNegativesRankingLoss`]() with these parameters:

```json
{
    "scale": 20.0,
    "similarity_fct": "cos_sim"
}
```

Evaluation Dataset

Unnamed Dataset

- Size: 31 evaluation samples
- Columns: `text1`, `text2`, and `label`
- Approximate statistics based on the first 1000 samples:

	text1	text2	label
type	string	string	int
details	• min: 5 tokens • mean: 8.39 tokens • max: 14 tokens	• min: 4 tokens • mean: 9.06 tokens • max: 14 tokens	• 1: 100.00%

- Samples:

text1	text2	label
花壇	花壇を調べよう	1
タイマツ要らない	キャンドル要らない	1
なにも要らない	欲しくない	1

- Loss: `MultipleNegativesRankingLoss` with these parameters:

```
{
    "scale": 20.0,
    "similarity_fct": "cos_sim"
}
```

Training Hyperparameters

Non-Default Hyperparameters

- `eval_strategy`: epoch
- `learning_rate`: 2e-05
- `num_train_epochs`: 13
- `warmup_ratio`: 0.1

- `fp16`: True
- `batch_sampler`: no_duplicates

All Hyperparameters

▶ Click to expand

Training Logs

Epoch	Step	Training Loss	loss	custom-arc-semantics-data_max_ap
None	0	-	-	1.0000
1.0	16	0.5617	0.5022	1.0000
2.0	32	0.2461	0.3870	1.0000
3.0	48	0.0968	0.3929	1.0000
4.0	64	0.0408	0.4012	1.0000
5.0	80	0.0151	0.4023	1.0000
6.0	96	0.0118	0.3851	1.0000
7.0	112	0.0087	0.3637	1.0000
8.0	128	0.0053	0.3662	1.0000
9.0	144	0.0046	0.3799	1.0000
10.0	160	0.002	0.3772	1.0000
11.0	176	0.0025	0.3765	1.0000
12.0	192	0.0021	0.3751	1.0000
13.0	208	0.0015	0.3752	1.0000

Framework Versions

- Python: 3.10.14
- Sentence Transformers: 3.0.1
- Transformers: 4.44.2
- PyTorch: 2.4.0+cu121
- Accelerate: 0.34.0
- Datasets: 2.20.0
- Tokenizers: 0.19.1

Citation

BibTeX

Sentence Transformers

```
@inproceedings{reimers-2019-sentence-bert,
    title = "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks",
    author = "Reimers, Nils and Gurevych, Iryna",
    booktitle = "Proceedings of the 2019 Conference on Empirical Methods in Natura
    month = "11",
    year = "2019",
    publisher = "Association for Computational Linguistics",
    url = "https://arxiv.org/abs/1908.10084",
}
```

MultipleNegativesRankingLoss

```
@misc{henderson2017efficient,
    title={Efficient Natural Language Response Suggestion for Smart Reply},
    author={Matthew Henderson and Rami Al-Rfou and Brian Strope and Yun-hsuan Sung
    year={2017},
    eprint={1705.00652},
    archivePrefix={arXiv},
    primaryClass={cs.CL}
```

```
}
```